ArohaSetu (आरोहासेतु)

"Notice Sooner, Care Better — A Personal Nurse for Every Family & Hospital Bed"

HACK FOR SOCIAL CAUSE 2027 REG: HSC|AN|00004

Viksit Bharat Young Leaders Dialogue (VBYLD 2027)

Knowledge Partner: IIT Bombay • **Organized by:** Ministry of Youth Affairs & Sports (MoYAS)

Primary Theme: Healthcare, Wellbeing & Service Delivery

Secondary Track: Elderly Care & Healthy Ageing • Assistive Tech

Target State / UT: Andaman & Nicobar Islands (Union Territory)

Live Prototype: https://arohasetu.vercel.app
Open Source: github.com/EternalFlames131/Arohasetu

Team & Institutional Identification

TEAM ID: HSC|AN|00004

Team Lead: Samrat Dasgupta (Teacher Trainee & Lead Innovator)

Role: Clinical workflow design, project idea, system architecture & regional validation

Team Member 2: Hriday Das (Main Developer / Coder)

Role: Full-stack implementation, edge vision pipeline, offline engine & testing

Team Member 3: C. Mohd. Jasheem (Designer)

Role: UI/UX design, visual experience, presentation & design assets

Institution: [College Name], Port Blair, Andaman & Nicobar Islands
AISHE Code: [AISHE Code to insert]

Problem Definition: The "Silent Gap" in Care

Vulnerabilities in Solo Elderly Care, Hospital Ward Overburden, and Island Geography

1. The Post-Fall "Long Lie"

30% of seniors fall annually. When an elder falls and cannot reach a phone for 6–12 hours, rapid rhabdomyolysis (muscle breakdown), pressure ulcers, hypothermia, and kidney failure set in.

Fatal Statistic: Over 50% of elders who experience a "long lie" do not survive past 6 months.

2. Silent Nocturnal Hypoxia

COPD, heart failure, and post-viral pulmonary patients suffer severe oxygen desaturation below 85% during deep sleep without waking up.

Hospital Crises: Night nurse-to-patient ratios exceed **1:30** in government wards, leaving hemodynamic deterioration unspotted between 4-hour manual rounds.

3. Andaman & Nicobar Island Reality

36 inhabited islands with only **one tertiary referral center (GB Pant Hospital, Port Blair)**. Outer islands (Hut Bay, Little Andaman, Car Nicobar) face 6–10 hour sea transit delays.

The Need: An offline-first, autonomous monitoring sentinel that detects deterioration immediately, without relying on stable internet or active panic buttons.

Why Existing Products Fail: Traditional panic buttons require the victim to be conscious to press them. Commercial smartwatches (Apple/Galaxy) suffer from 80%+ false alarms during hand gestures, cost ₹30,000–₹80,000, and cannot be afforded by rural families.

Proposed Solution: The Arohasetu Architecture

Two-Layer Safety Shield + Autonomous 3-Tier Escalation Ladder + Dual Deployment

Layer 1: Wearable Physiological Shield

- Ingests vital biometrics: **Heart Rate, SpO2, Blood Pressure, Temperature, Respiratory Rate, Blood Glucose**.
- Analyzes trends via the clinically proven **NEWS2** (National Early Warning Score 2) deterioration matrix.
- Replaces black-box generative AI with auditable, deterministic clinical rules to eliminate medical hallucinations.

Layer 2: Privacy-First Optical Sentinel

- Uses an overhead commodity webcam running on-device edge AI (**Ultralytics YOLO-Pose**).
- Extracts 17 skeletal joint coordinates at 25+ FPS to detect sudden falls, floor stillness, and wandering.
- **100% Video-Private:** Zero video is recorded, stored, or streamed to the cloud (DPDP Act 2023 compliant).

Autonomous 3-Tier Fail-Safe Escalation Ladder

Tier 1 (0–15s): Instant push notification and automated voice call to primary family caregiver.

Tier 2 (15–45s): Automated escalation to secondary contact, neighbor, or attending physician if unacknowledged.

Tier 3 (45s+): Autonomous emergency dispatch (108 / 112 / Island Marine Ambulance) with telemetry payload.

Technical Implementation & Working Prototype

A Fully Operational, Production-Deployed Software Ecosystem (Not a Figma Mockup)

WEB & CLOUD

React 18 + Node.js

Serverless API on Vercel (`arohasetu.vercel.app`) with HMAC-SHA256 authenticated sessions, sub-1.8s live sync, and responsive PWA layout.

EDGE COMPUTER VISION

YOLO-Pose Daemon

Python sentinel extracting 17 COCO keypoints locally with GPU/CPU acceleration. Dedicated capture worker delivers 9.6ms fresh snapshots.

MOBILE ECOSYSTEM

Native Android App

Standalone Kotlin + Jetpack Compose APK (`Download/Arohasetu-Demo.apk`) running 100% offline with embedded rules engine and MedStore.

EVALUATOR TESTBED

Admin Simulator

Developer cockpit at `/admin/simulator` for judges to trigger clinical presets (High HR, Low SpO2, BP Crisis, Disconnect) with live sync.

Dual Deployment Modes

Home Caregiver Portal: Complete 5-language localization (English, Hindi, Bengali, Tamil, Telugu), Prajñā AI Assistant, medication adherence tracker, and privacy-first kinematic radar.

Hospital "Virtual Ward": Multi-bed telemetry grid (Beds 101–104) with dynamic NEWS2 acuity sorting, 4-stage clinician lifecycle (`New` → `Acknowledged` → `Follow-up` → `Resolved`), and ABDM audit trails.

Social Impact, Scalability & Healthcare Integration

Democratizing ICU-Grade Monitoring for 140M+ Indian Elders and Rural Primary Centers

1. National Health Alignment

- **Ayushman Bharat (ABDM):** Designed for 14-digit ABHA Health ID integration and Electronic Health Record export.
- **Tele-MANAS & 108 Emergency:** Integrated with national emergency helpline dispatch protocols.
- **UN SDGs:** Directly addresses **SDG 3** (Good Health & Well-being) and **SDG 10** (Reduced Inequalities).

2. Frugal Social Innovation

- **Replaces ₹3,00,000 ICU Beds:** Software-first approach converts any ₹800 webcam and existing smartphone into an optical monitor.
- **Hardware Roadmap:** Indigenous "Made in India" multi-sensor IoT bedside gateway planned at <₹5,000 for rural Primary Health Centres.
- **Saves Rural Families:** Prevents emergency ICU admissions caused by undetected falls and delayed interventions.

3. Inclusive Vernacular Access

- **5 Indian Languages:** English, Hindi (हिन्दी), Bengali (বাংলা), Tamil (தமிழ்), Telugu (తెలుగు).
- **Empowers ASHA / ANMs:** Simple color-coded triage allows community health workers to monitor rural sub-centers.
- **Island Resilience:** Works during cellular or submarine fiber cuts via local edge processing.

The Viksit Bharat 2047 Vision: Ensuring that no senior citizen in India—whether living in a metro apartment in Delhi or an isolated wooden house in Little Andaman—ever dies alone and undiscovered during a preventable medical crisis.

Clinical Reliability, Validation & Prototype Honesty

7 Built-in Failsafes, 100% Automated Regression Pass Rate, and Strict Academic Integrity

7 Built-in Clinical Reliability Safeguards

- **1. Physiological Range Validation:** Filters impossible sensor artifact spikes.
- **2. Device Confidence Indexing:** Rates sensor skin contact (0–100%).
- **3. Consecutive Verification:** Requires sustained danger before EMS dispatch.
- **4. Sensor Heartbeat Watchdog:** Instantly alerts if a sensor powers off or drops.
- **5. Alarm Rate-Limiting:** Groups repetitive alerts to eliminate alarm fatigue.
- **6. Immutable ABDM Audit Trail:** Logs every clinician note with timestamps.
- **7. Multi-Modal Graceful Degradation:** Vision watch persists if wearables die.

Exhaustive Automated Regression Testing

- **23/23 False-Positive Scenarios Passed (100%):** Tested sitting fast, tying shoelaces, dropping phones, lying on sofa, and furniture occlusions.
- **7/7 Kinematics Unit Tests Passed (100%):** Verified velocity derivatives ($\frac{dv}{dt}$), torso inclination angle (θ), and ground stillness windows.

Prototype Honesty Disclosure:

• **REAL:** Web application, Virtual Ward, NEWS2 rules engine, local YOLO-Pose vision, offline Android app, 4-stage nurse lifecycle.

• **SIMULATED:** Physical BLE hardware ingestion (via Admin Simulator) and carrier telephone dialing to 108/112 (simulated for live demo safety).

Roadmap, Team Synergy & Mandatory AI Disclosure

Phased Execution Plan, Community-Rooted Leadership, and Transparent Tool Usage

Phased Execution Roadmap (2026–2027)

- **Phase 1 (Completed - Oct 2026):** Production web portal, local YOLO edge sentinel, offline Android APK, and simulator verified for HSC 2027.
- **Phase 2 (Nov–Dec 2026):** Grassroots pilot with ASHA workers in Port Blair and hardware BLE sensor testing.
- **Phase 3 (Jan 2027):** National showcase demonstration at VBYLD New Delhi (IIT Bombay); pilot at GB Pant Hospital Virtual Ward.
- **Phase 4 (2027+):** Integration with Ayushman Bharat ABHA IDs and low-cost indigenous bedside IoT gateway manufacturing (<₹5,000).

Official Mandatory AI-Use Disclosure

Human Ownership & Authorship: The core concept, healthcare problem definition, two-layer architecture, clinical workflows, UI/UX designs, and testing scenarios were entirely conceived, self-made, and created by the team members.

Tools Utilized: AI Coding Assistant (Generative AI Developer Tools).

Scope of Usage: AI was used strictly to do light bug checks and light coding only; other than that, everything was self-made and created entirely by the team.

"ArohaSetu: Notice sooner, care better — Empowering families and safeguarding lives across Bharat."